

UNIVERSITY OF CALIFORNIA

SANTA CRUZ

**Cross-Query Optimization for
Worst-Case Optimal Joins in Apache Calcite**

A project submitted in satisfaction
of the requirements for the degree of

MASTER OF SCIENCE

in

COMPUTER SCIENCE & ENGINEERING

by

Thomas Banghart

March 2026

The Master's Project of
Thomas Banghart is approved:

Professor Nikos Tziavelis, Chair

Professor Ioannis Demertzis, Reader